Association Between Patient Characteristics and Low-Dose Naltrexone Use Among Palliative Care Patients

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Background

Low-dose naltrexone (LDN; ≤4.5 mg/day) is used off-label as a non-opioid option for chronic pain.

Recent placebo-controlled trials show no clear benefit, and its use in palliative care has never been described at the population level.

LDN at a glance

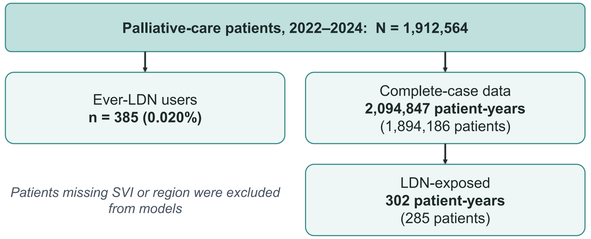
- Naltrexone is approved at ≥50 mg for alcohol and opioid use disorders
- Only 50-mg tablets are sold, so 1–5 mg doses are usually compounded
- Medicare Part D covers only the covered components of a compound

Objective

Estimate trends in recorded LDN use among U.S. patients receiving palliative care (2022–2024) and the patient factors associated with use.

Methods

- **Data:** Epic Cosmos, de-identified EHR from >330 health systems
- **Cohort:** U.S. residents meeting palliative-care criteria (Z51.5/V66.7 or palliative-specialist consult codes)
- **Exposure:** naltrexone order ≤4.5 mg/day or a compounded low-dose formulation
- **Analysis:** patient-year logistic regression, patient-clustered SEs; adjusted for age, sex, race and ethnicity, insurance, neighborhood SVI, region, and year

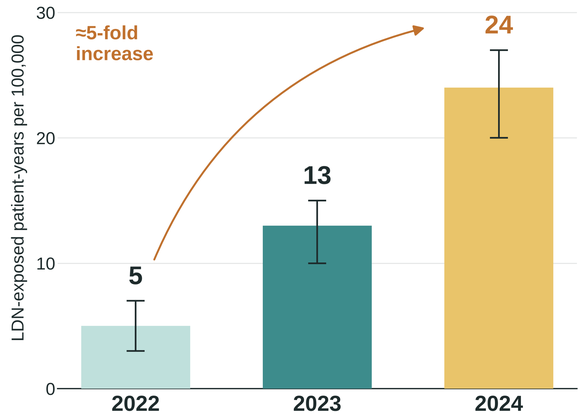


Patients missing SVI or region were excluded from models

KEY FINDING Recorded LDN use rose about **5-fold** from 2022 to 2024 but stayed **rare** (under 3 per 10,000 patient-years), and differed by age, sex, race and ethnicity, insurance, neighborhood, and region.

1.9 M	385	14.4	4.97×
palliative-care patients	ever-LDN users (0.020%)	per 100,000 patient-years	adjusted odds, 2024 vs 2022

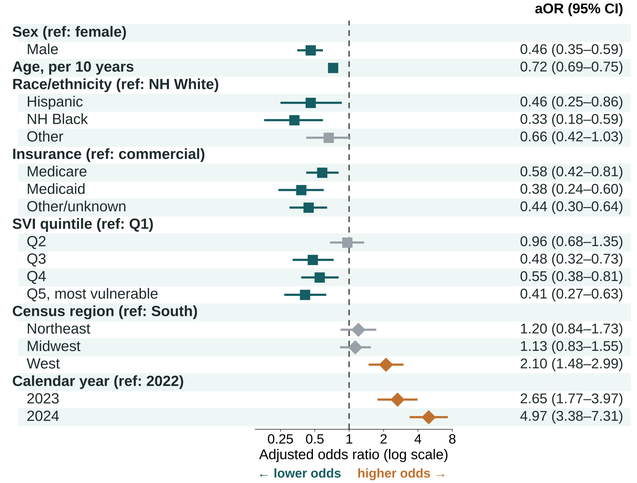
Adjusted use rose every year



Adjusted predicted probability (95% CI), marginal standardization: 0.005%, 0.013%, 0.024%.

Each added decade of age: 28% lower odds of LDN use (OR 0.72, 95% CI 0.69–0.75)

Who receives LDN? Adjusted odds



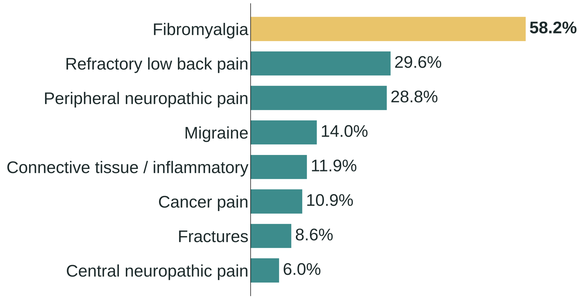
2,094,847 complete-case patient-years (302 LDN-exposed); cluster-robust CIs; mutually adjusted. Grey = CI crosses 1. NH, non-Hispanic; SVI, Social Vulnerability Index (Q1 = least vulnerable).

LDN users were younger, more often female, and more often commercially insured

	Age, mean	Female	NH White	Commercial	Medicare	West	SVI Q1–Q2
Users (n = 385)	**61.3 y**	**67.3%**	**84.2%**	**30.6%**	**48.1%**	**19.2%**	**48.0%**
Non-users	74.2 y	51.6%	73.0%	10.7%	64.4%	11.7%	29.7%

Patient-level, unadjusted (column %). SVI Q1–Q2 = two least socially vulnerable neighborhood quintiles.

Fibromyalgia in 58% of users



Prespecified conditions in the 365 days before the first LDN record (n = 385); not mutually exclusive; users only.

Discussion

- A large relative rise in a rare exposure, while trial evidence for LDN did not strengthen
- User profile (fibromyalgia, back and neuropathic pain) suggests chronic non-malignant pain
- For older adults, indication, benefit, and opioid interactions remain unknown
- Differences reflect recorded use, not measured need or appropriateness

Limitations

- Medication orders, not dispensing or use
- Palliative-care codes not validated in outpatient EHR data
- Rare outcome: imprecise subgroup estimates

Take-home

LDN use in palliative care is rare but rising despite weak evidence. Next: opioid co-prescribing (MME), indication, dispensing, and outcomes.

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Key references

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